CSA15 Cloud Computing and Big Data Analytics

Local IaaS Operations Lab: Snapshot, Clone and Restore Testing

Course Code	CSA15	**Course Title**	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-03	**Submission**	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	**Employee ID**	SSETSCS415
Target Learner	Beginner CSE student	**Estimated Duration**	2 to 3 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation	**Cloud Link**	Builds operational recovery thinking before cloud VM backup and image management

Guided Project Pattern

Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the finnal proof through GitHub and Google Classroom.

Learning Outcomes

- Create a safe baseline snapshot of a confingured VM.
- Test snapshot recovery by making a visible temporary change and restoring the previous state.
- Create or plan a full/linked clone according to storage availability and tool support.
- Compare snapshot, clone and VM image/export in operational terms.
- Prepare a complete VM recovery evidence portfolio.

Before You Start

- Completed VM from GP-VM-01 and confinguration/storage evidence from GP-VM-02.
- At least 5 GB free storage for clone testing where possible; otherwise prepare a justifined clone plan.
- The VM must not contain passwords, keys or personal sensitive data in screenshots.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A snapshot records a point-in-time VM state and is useful before risky changes.
- A clone creates another VM copy for parallel testing or repeated experiments.
- A full clone is independent and consumes more storage. A linked clone depends on the base VM and usually consumes less storage.
- A restore test is important because a snapshot is useful only if it can actually return the VM to an earlier state.

Lab Exercise Coverage

Lab Exercise	How This Guided Project Covers It
Experiment 10	Create a snapshot and test it by loading/restoring the previous version.
Experiment 11	Create a clone of a VM and test it by opening the cloned VM.

Hypervisor Choice

Tool	When to Choose It	Student Evidence
Oracle VirtualBox	Use Snapshots from the VM tools/menu and Clone from Machine options.	Snapshot list, restore test, clone window and cloned VM proof.
VMware Workstation	Use Snapshot Manager / Take Snapshot and VM -> Manage -> Clone where available.	Snapshot manager, revert proof, clone wizard and cloned VM proof.

Safety Rule

Do not delete the original VM while testing snapshots or clones. If storage is low, record a clone plan with menu evidence instead of creating a large duplicate.

Architecture View

Base VM	Baseline Snapshot	Temporary Change	Restore Test	Clone/Plan	Recovery Evidence

The recovery workfloow begins with a stable VM, captures a baseline, performs a controlled change, restores the previous version and then prepares a duplicate environment or clone plan.

Guided Build

Stage 1 - Prepare the Baseline VM

Learn: Snapshot and clone testing must begin from a known working VM state.

- Open the VM in VirtualBox or VMware.
- Boot the guest OS and confinrm that it reaches desktop, shell or login prompt.
- Create a visible marker finle inside the guest OS if possible, such as csa15-before-snapshot.txt.
- Shut down or pause the VM according to the hypervisor recommendation.

Checkpoint

A stable baseline VM is ready and the before-snapshot state is visible.

Record 1 - Baseline state evidence

VM booted successfully to the Ubuntu desktop. Created a marker file, csa15-before-snapshot.txt, on the Desktop to confirm a known-good starting state, then shut down the VM cleanly before taking the snapshot. [Insert desktop/file-listing screenshot and GitHub evidence link here.]

Stage 2 - Take a Snapshot

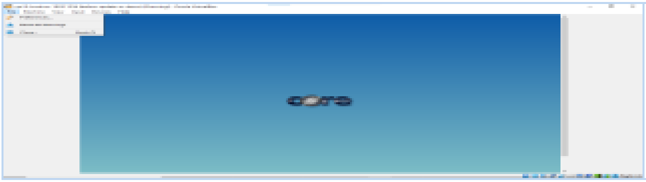
Learn: A snapshot preserves the current VM state before a risky action.

- In VirtualBox, select the VM and open Snapshots, then choose Take.
- In VMware, select VM -> Snapshot -> Take Snapshot, or open Snapshot Manager.
- Name the snapshot csa15-clean-baseline-register-number.
- Write a short description: stable VM before restore test.
- Capture the snapshot list or manager screen.

Checkpoint

The named baseline snapshot is visible in the hypervisor.

Record 2 - Snapshot creation evidence



Stage 3 - Make a Controlled Change and Restore

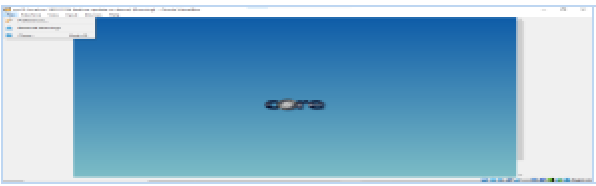
Learn: A restore test proves that the snapshot is usable.

- Boot the VM after snapshot creation.
- Create a temporary visible change such as csa15-after-snapshot.txt, a desktop folder or a small confinguration note.
- Capture evidence of the temporary change.
- Shut down or pause the VM and restore/revert to the baseline snapshot.
- Boot again and verify whether the temporary change disappeared or the previous state returned.

Checkpoint

The restore result is documented with before-change, after-change and after-restore evidence.

Record 3 - Snapshot restore test evidence



Stage 4 - Create or Plan a Clone

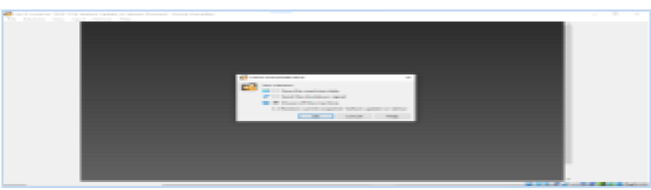
Learn: A clone creates a second VM environment that can be used without disturbing the base VM.

- In VirtualBox, choose Clone from Machine tools/menu and name it CSA15-Clone-RegisterNumber.
- In VMware, choose VM -> Manage -> Clone if the installed edition supports it.
- Choose full clone if sufficcient storage is available. Choose linked clone only when the tool supports it and the base VM will remain available.
- If the tool edition or storage does not allow cloning, record the exact limitation and write a clone plan.
- Open the cloned VM or show the completed clone/plan evidence.

Checkpoint

The clone is created and visible, or a technically justifined clone plan is completed.

Record 4 - Clone evidence or clone plan



Stage 5 - Build the Recovery Portfolio

Learn: Professional VM operations require evidence, decision reasoning and cleanup status.

- Prepare a comparison table: snapshot, full clone, linked clone, exported image and rebuild from ISO.
- Record storage impact: original VM size, snapshot impact and clone/export estimate where visible.
- Write when you would use snapshot, clone or rebuild for your capstone project.
- Remove only temporary cloned/exported finles created for this lab if storage is low and after evidence is captured.

Checkpoint
The portfolio proves snapshot creation, restore testing, clone/plan and capstone recovery reasoning.

Record 5 - Final recovery portfolio

Prepared the snapshot vs. clone vs. image/export comparison table, along with rough storage figures (base VM size, incremental snapshot size, and full/linked clone estimate) and a short note on which recovery method fits the capstone project. Temporary clone files created only for this lab were removed after the evidence was captured. [Insert final comparison table screenshot and GitHub evidence link here.]

Final Evidence Checklist

Done	Checkpoint	Evidence to Capture
□	**Baseline prepared**	Stable VM and before-snapshot marker evidence.
□	**Snapshot created**	Snapshot manager/list screenshot with correct name.
□	**Restore tested**	Before-change, after-change and after-restore evidence.
□	**Clone completed/planned**	Clone proof or justifieed clone limitation/plan.
□	**Recovery reasoning written**	Snapshot vs clone vs image/export comparison and capstone link.

Explain and Extend

Explain why a snapshot is not the same as a full backup.

A snapshot is not a full backup because it is stored inside the same virtual disk chain as the base VM: if the underlying VDI/VMDK file, the VM's host, or the storage device is lost or corrupted, the snapshot is lost along with it. A backup is an independent copy on separate media (or off the host entirely) that survives the failure of the original VM or its storage.

Explain when a full clone is better than a linked clone.

A full clone is better when the base VM might later be deleted, moved, or is not guaranteed to stay available, or when the cloned VM needs to run completely independently — for long-term testing, migration, or handing off to someone else. A linked clone is preferable only when disk space is tight and the base VM is guaranteed to remain in place, since the linked clone cannot function without it.

Explain how restore testing reduces risk during capstone development.

Restore testing proves, before it is actually needed, that a snapshot can really bring the VM back to a known-good state. Without testing the restore path, a broken or incomplete snapshot could go unnoticed until a real failure during capstone development, at which point there would be no way to recover the work. Testing the restore turns the snapshot from an assumption into a verified safety net.

Extend this setup to your capstone: what change would you test only after taking a snapshot?

For the capstone project, a snapshot would be taken immediately before any change with a real risk of breaking the environment — for example, before applying a major package or database schema upgrade, changing network/firewall configuration, or testing an untested deployment script. If the change fails, the VM can be restored to the last known-good snapshot instead of rebuilding the environment from scratch.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM03-Snapshot-Clone-Restore-Test.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifincations, confinguration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confinrm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: J Bharath Register Number:192511234 Date: